Estimating the Net Present Value of a New Credit Card Customer:
A Literature Survey and Modeling Framework for Retail Banks

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Abstract

Retail banks use credit card campaigns to acquire new customers, deepen existing relationships, and shift payment activity onto proprietary products. The commercial objective is not simply response, approval, activation, or first purchase. It is incremental risk-adjusted net present value (NPV), net of campaign cost, rewards, servicing, funding, capital, fraud, expected credit loss, and cannibalization of the bank's existing products. This survey reviews the literatures most relevant to that decision problem. Household-finance and credit-card studies show that spending and borrowing are state dependent: employment, income shocks, liquidity, credit limits, interest rates, policy transfers, and macro shocks all alter card economics. Payment-choice and rewards studies show that adoption and use are distinct outcomes, so a newly issued card may remain inactive, cannibalize existing spend, or capture incremental wallet share. Customer lifetime value and duration models provide useful baselines for forecasting transaction frequency, monetary value, active status, and relationship duration, but they must be integrated with credit risk and balance-sheet economics for card portfolios. The paper proposes a modular modeling architecture for estimating incremental credit-card NPV and identifies where public literature is informative and where issuer-specific randomized or quasi-experimental evidence is essential.

1 Introduction

A large U.S. retail bank considering a credit card campaign faces two related business cases. First, the bank may market a card to a prospect who is not yet a card customer. Second, it may offer a new card to an existing client, including an existing bank customer with deposits or loans, or an existing cardholder who could add another card. In both cases, the operational decision is often framed as a propensity problem: who will respond, who will be approved, who will activate, and who will spend. That framing is useful for funnel management, but it is incomplete for capital allocation. The economically relevant object is a profit or value objective, not a response score alone. This is consistent with the direct-marketing literature on optimal selection, which ranks customers by expected net return rather than by response probability alone [10], and with the customer lifetime value (CLV) and customer-equity literatures, which define customer value as a discounted stream of expected future contribution [5, 29, 33]. The credit-card setting adds bank-specific components—losses, capital, funding, fraud, rewards, and cannibalization—to this familiar marketing objective.

Let i index customers, t monthly time, and $a \in \mathcal{A}$ a marketing or product action, such as a direct-mail offer, digital preapproval, branch offer, balance-transfer campaign, rewards bonus, or product upgrade. In the direct-mail formulation, the bank would mail customer i when expected margin times response probability exceeds contact cost:

$$d_i^* = \mathbf{1}\{p_i m_i - c_i > 0\}, \tag{1}$$

where p_i is response probability, m_i is conditional gross margin, and c_i is contact cost. In a CLV formulation, customer value is instead a discounted contribution stream:

$$\text{CLV}_i = \mathbb{E}\left[\sum_{t=0}^{T} \delta_t \{R_{it} - C_{it}\} \,\middle|\, \mathcal{I}_{i0}\right]. \tag{2}$$

The card-acquisition problem combines these ideas with treatment effects:

$$\Delta\text{NPV}_i(a) = \mathbb{E}\left[\sum_{t=0}^{T} \delta_t \left\{\Delta R_{it}(a) - \Delta C_{it}(a) - \Delta L_{it}(a) - \Delta K_{it}(a)\right\} - A_i(a) \,\middle|\, \mathcal{I}_{i0}\right], \tag{3}$$

where ΔR_{it} is incremental revenue, ΔC_{it} is incremental servicing, rewards, fraud, and funding cost, ΔL_{it} is expected credit loss net of recovery, ΔK_{it} is capital or balance-sheet cost, $A_i(a)$ is acquisition and setup cost, and $\mathcal{I}_{i0}$ is information available at the decision date. The increments are measured against the counterfactual in which the customer does not receive action a.

Thus, Equation (3) is not claimed to be a single canonical equation from one credit-card paper. It is the synthesis needed for a bank: Equation (1) contributes the expected-profit targeting logic, Equation (2) contributes the discounted lifetime stream, and causal/uplift methods contribute the requirement that all terms be incremental to the no-offer counterfactual.

This counterfactual is crucial. For an existing client, observed post-campaign spend on a new card can be high even when incremental NPV is low: the card may shift spend from the bank's old card, pull forward spend through a costly promotion, or create balances whose losses exceed finance charges. Conversely, a moderate activation probability can be valuable if it creates durable incremental purchase volume, profitable revolving balances, and low servicing and loss costs. The bank's literature problem is therefore not a search for a single "credit-card NPV paper." It is a synthesis problem across household finance, payment choice, credit-card economics, causal marketing, and customer lifetime value.

This paper is organized around three questions. Section 2 reviews evidence on how market conditions, employment, income, and liquidity alter spending and card economics. Section 3 reviews adoption, activation, rewards, and use of new payment instruments. Section 4 reviews customer lifetime value and duration models for lifetime expenditure. Section 5 proposes a practical model architecture for a bank. Section 6 develops a component decomposition into losses, balances, PPNR, and portfolio-management overlays. Section 7 addresses measurement and decision pitfalls that arise when NPV is used across the card funnel. Section 8 discusses empirical design and validation. Section 10 records the source-support limits of this first-pass survey.

2 Market Conditions, Employment, and Propensity to Spend

2.1 Income, unemployment, and liquidity

The first component of credit-card NPV is the customer's capacity and propensity to spend. The relevant empirical papers do not estimate the bank's Equation (3) directly. They estimate causal or quasi-causal effects of income, employment, and liquidity variables on spending or debt. Their contribution to our model is a set of state variables and elasticities for the spend, balance, and risk modules.

Ganong and Noel [15] use de-identified bank account data to study spending around unemployment and the predictable income drop at unemployment insurance exhaustion. A stylized event-study version of the object is

$$y_{i\tau} = \alpha_i + \lambda_\tau + \sum_{k \neq -1} \beta_k \mathbf{1}\{\tau_i = k\} + \varepsilon_{i\tau}, \tag{4}$$

where $y_{i\tau}$ is spending for household i at event time τ relative to an unemployment or benefit-exhaustion event, α_i absorbs household fixed effects, and the β_k trace the spending path around the event. The important modeling implication is not merely "unemployment matters." It is that the card spend forecast should include event-time variables such as job loss, payroll interruption, benefit receipt, and benefit exhaustion, because spend can move non-smoothly around predictable income events.

Johnson et al. [19] study the 2001 tax rebates using variation in rebate timing. A stylized specification is

$$\Delta c_{it} = \alpha_i + \lambda_t + \gamma \, \mathrm{Rebate}_{it} + \eta' H_i \, \mathrm{Rebate}_{it} + u_{it}, \tag{5}$$

where γ is a marginal propensity to consume out of the transfer and H_i contains heterogeneity variables such as income or liquid-wealth status. For a credit-card model, this says that transfers, payroll changes, and liquid wealth proxies should enter monthly purchase-volume and payment-rate equations, with heterogeneous effects. It does not identify the share of extra consumption that will route through our card.

Gross and Souleles [16] provide the closest direct credit-card evidence. They study account-level card and bureau data and estimate how debt responds to credit limits and interest rates. A stylized account-level equation is

$$\Delta B_{it} = \alpha_i + \lambda_t + \theta \, \Delta \mathrm{Limit}_{it} + \rho \, \Delta \mathrm{APR}_{it} + \psi' X_{it} + e_{it}, \tag{6}$$

with stronger responses among initially high-utilization or liquidity-constrained borrowers. This should enter our future model through the balance and payment modules: line assignment and APR are not passive controls; they are treatment components that affect revolving balances, utilization, finance charges, and expected loss.

2.2 Rates, regulation, and product terms

Credit-card NPV is also shaped by prices, fees, disclosures, repayment rules, and regulatory constraints. Agarwal et al. [2] study the Credit Card Accountability Responsibility and Disclosure Act using a large account-level card panel. The paper is not a spend-propensity model, and this survey should not use it as if it supplied a universal activation or balance elasticity. Its modeling value is different: it shows that product terms are governed economic objects. Fees, disclosures, and repayment nudges can change both issuer revenue components and consumer repayment behavior, so the NPV model cannot treat APRs, fees, and promotional terms as free scalar revenue add-ons.

Formally, the campaign action should be represented as a vector of product terms, not as a binary contact flag:

$$a_{it} = \left(\ell_{it}, r_{it}^{\mathrm{purchase}}, r_{it}^{\mathrm{cash}}, r_{it}^{\mathrm{BT}}, r_{it}^{\mathrm{promo}}, T_{it}^{\mathrm{promo}}, f_{it}, q_{it}, b_{it}, d_{it} \right) \in \mathcal{A}_{it}(X_{it}, M_{gt}, G_t). \tag{7}$$

Here ℓ is the credit line; the r terms are purchase, cash-advance, balance-transfer, and promotional APRs; T^{promo} is promotional duration; f is the vector of annual, late, balance-transfer, cash-advance, foreign-transaction, and waiver terms; q is the reward earn or redemption schedule; b is a signup or statement-credit bonus; and d is the disclosure, minimum-payment, autopay, and digital-wallet provisioning design. The feasible set $\mathcal{A}_{it}$ is indexed by customer state, macro state, and governance state G_t because underwriting policy, compliance rules, disclosure requirements, risk appetite, pricing floors, and operational constraints limit which offers can be made.

This notation connects the CARD Act evidence to the later NPV architecture in two ways. First, regulation changes the feasible set and sometimes the cash-flow mapping:

$$a_{it} \in \mathcal{A}_{it}(G_t), \tag{8}$$

$$CF_{i,t+1} = c(Z_{it}, X_{it}, M_{gt}, a_{it}; G_t). \tag{9}$$

A rule change, fee cap, disclosure change, or repayment-allocation rule is therefore not merely an external dummy variable. It changes either the offers available to the bank or the function that maps behavior into interest, fees, payments, and balances. Second, product terms are treatments that enter the behavioral transition system. A reduced-form version is

$$\Pr(Z_{i,t+1} = z' \mid Z_{it}, X_{it}, M_{gt}, a_{it}, G_t), \tag{10}$$

$$\mathbb{E}[Y_{i,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a_{it}, G_t], \qquad Y \in \{S, B, Pay, D, Q\}. \tag{11}$$

Thus APR, line, fees, rewards, and disclosures can affect activation, purchase volume, revolving balances, payments, delinquency, rewards liability, servicing cost, and attrition. They should not be appended only after a response model has already selected customers.

The mathematical bridge to Gross and Souleles [16] is direct. Equation (6) says that limit and APR changes can move card debt, especially for liquidity-constrained accounts. In the NPV engine, the same terms must feed both revenue and risk:

$$B_{i,t+1} = g_B(B_{it}, Pay_{it}, S_{it}, \ell_{it}, r_{it}, f_{it}, X_{it}, M_{gt}) + \varepsilon^B_{i,t+1}, \tag{12}$$

$$Int_{i,t+1} = r^{\text{revolve}}_{it} B^{\text{revolve}}_{i,t+1} + r^{\text{cash}}_{it} B^{\text{cash}}_{i,t+1} + r^{\text{BT}}_{it} B^{\text{BT}}_{i,t+1}, \tag{13}$$

$$EL_{i,t+1} = PD_{i,t+1}(Z_{it}, U_{it}, X_{it}, M_{gt}, a_{it})\, LGD_{i,t+1}\, EAD_{i,t+1}(B_{i,t+1}, \ell_{it}, a_{it}). \tag{14}$$

Raising APR can mechanically increase the rate applied to revolving balances, but it can also change borrowing, payment, attrition, and default behavior. Increasing the line can raise usage and finance revenue, but it can also raise utilization dynamics, exposure at default, capital, and downside risk. The correct object for decisioning is therefore a total derivative of value with respect to the governed offer vector, not a partial revenue calculation holding behavior fixed. Let $\mu_Y(a) = \mathbb{E}[Y_i(a) \mid \mathcal{I}_{it}]$ for $Y \in \{Z, S, B, Pay, D, Q\}$. For any feasible local perturbation v of the offer vector,

$$\begin{aligned} D_v\mathbb{E}[\Delta\text{NPV}_i(a)] = {} & \nabla_a \mathbb{E}[\Delta\text{NPV}_i(a, \mu)]^\top v \Big|_{\mu=\mu(a)} \\ & + \sum_{Y \in \{Z,S,B,Pay,D,Q\}} \frac{\partial \mathbb{E}[\Delta\text{NPV}_i(a, \mu)]}{\partial \mu_Y}\Bigg|_{\mu=\mu(a)} D_v \mu_Y(a). \end{aligned} \tag{15}$$

The first term is the mechanical cash-flow effect of the term sheet; the second term is the behavioral response channel. The second term is precisely why the rates and regulation literature belongs before the usage, CLV, loss, balance, and PPNR sections.

This also clarifies how product terms should be validated. The promotion criterion is not whether a high-APR or high-fee offer raises modeled revenue conditional on the same balance path. It is whether the feasible offer $a \in \mathcal{A}_{it}(G_t)$ increases counterfactual value after behavioral response, regulation, cannibalization, expected loss, capital, funding, rewards, and servicing:

$$\Delta\mathrm{NPV}_i(a;G_t) = -\Delta A_i(a) + \sum_{h=0}^{H} \delta_h\, \mathbb{E}\left[\begin{aligned}&\Delta PPNR_{i,t+h}(a;G_t) - \Delta EL_{i,t+h}(a;G_t) - \Delta K_{i,t+h}(a;G_t)\\ &\quad + \Delta RelValue_{i,t+h}(a) \mid \mathcal{I}_{it}\end{aligned}\right]. \tag{16}$$

Equation (16) is the handoff to the component decomposition in Section 6: promotional APRs and balance-transfer terms become promo states and balance subledgers; fees and waivers become non-interest income; rewards become expense and liability; APR/line choices affect balances, EAD, and capital; disclosure and repayment design enter payment and attrition transitions; and all terms remain subject to the feasible governed action set.

2.3 Macro shocks and category substitution

Market shocks can change not only the level of spending but also the composition of card spending. Baker et al. [3] use transaction-level household financial data to study consumption during the COVID-19 shock. Their evidence documents large changes in spending levels and categories during the epidemic. For card NPV, category shifts matter because merchant-category mix affects interchange, rewards expense, fraud risk, dispute rates, and charge-off correlation. A recession, health shock, inflation shock, or local labor-market shock can lower discretionary purchase volume while increasing borrowing demand and credit risk.

The practical lesson is that spend models should be time varying and category-aware. A customer with high pre-campaign grocery and travel spend is not equally valuable in every macro state. A useful reduced-form component is

$$\log(1 + S_{ij,t+1}) = \mu_i + \lambda_t + \kappa_j + \beta_j' M_{gt} + \chi_j' X_{it} + \omega_j'(M_{gt} \times X_{it}) + \epsilon_{ij,t+1}, \tag{17}$$

where $S_{ij,t+1}$ is spend in merchant category j, M_{gt} is a local or national macro vector for geography g, and X_{it} includes customer liquidity, payroll, utilization, and relationship variables. Equation (17) is not lifted from one paper; it is the model slot suggested by the income, liquidity, and macro-shock evidence.

3 Will the Newly Issued Card Be Used?

3.1 Adoption is not usage

The second requested problem is the probability that a customer who receives a new card will use that card. The payment-choice literature warns against collapsing this into a single response probability. Koulayev et al. [20] estimate a structural model of adoption and use of payment instruments by U.S. consumers. Their central relevance here is that possession of a payment instrument and use of that instrument are distinct choices. A customer may open a card, activate it, and still use another issuer's card, debit card, cash, ACH, or a digital wallet at the point of sale.

Mathematically, the useful structure is a two-stage adoption/use problem. A stylized version is:

$$A^*_{im} = Z_i'\alpha_m + \nu_{im}, \qquad A_{im} = \mathbf{1}\{A^*_{im} > 0\}, \tag{18}$$

$$U_{ijm} = X_{ijm}'\beta_m + \phi_m A_{im} + \varepsilon_{ijm}, \qquad m^*_{ij} = \arg\max_{m \in \mathcal{M}_i} U_{ijm}. \tag{19}$$

Here A_{im} denotes whether consumer i has adopted payment instrument m, U_{ijm} is the utility of using instrument m for transaction context j, and $\mathcal{M}_i$ is the set of adopted instruments available

Table 1: How Section 2 evidence should enter a bank model.

Literature	Mathematical object	Model use
Ganong and Noel [15]	Event-time spending path around unemployment and benefit exhaustion.	Add job-loss, payroll interruption, benefit receipt, and exhaustion indicators to spend, payment, attrition, and loss modules.
Johnson et al. [19]	Marginal propensity to consume out of transfer timing, with heterogeneity.	Add income/transfer shocks and liquidity interactions to spend and payment equations; do not assume routing to our card.
Gross and Souleles [16]	Account-level response of card debt to credit limits and APR.	Treat line and APR as treatment components affecting balances, finance revenue, utilization, and loss.
Agarwal et al. [2]	Regulation-induced changes in fees, disclosures, repayment behavior, and issuer revenue components.	Treat APRs, fees, rewards, disclosures, and repayment rules as governed terms $a_{it} \in \mathcal{A}_{it}(G_t)$ that feed transition, balance, loss, PPNR, and promo-state equations rather than unconstrained revenue levers.
Baker et al. [3]	Transaction-level response of spending levels and categories to a macro shock.	Forecast category-level spend and interchange/rewards/fraud mix under macro scenarios.

to consumer i. The exact specification in Koulayev et al. [20] is richer, but Equations (18)–(19) capture the modeling lesson for us: "has the card" gates the choice set, while "uses the card" is a conditional payment-choice decision. Our card model should therefore estimate activation and usage conditional on the customer's competing payment portfolio rather than treating issuance as usage.

For a credit-card campaign, the operational funnel should therefore separate:

1. offer exposure and response;
2. application and approval;
3. card opening and physical or digital activation;
4. first purchase;
5. repeated purchase;
6. share-of-wallet capture;
7. durable active state, dormancy, closure, or charge-off.

Each transition has different covariates and different economic meaning. Approval is partly a risk and underwriting outcome. Activation is partly a friction and onboarding outcome. First purchase is an early usage outcome. Share-of-wallet is a competitive payment-routing outcome. Durable activity is a lifetime-value outcome.

3.2 Rewards, reactivation, and early spend

Rewards can alter usage, but the bank must distinguish profitable use from costly promotional pull-forward. Agarwal et al. [1] study cash-back rewards using administrative data from a large U.S. financial institution. The important point for our later model is that the paper is not just a qualitative statement that "rewards matter." It suggests three account-level response objects:

$$A_{it}^{\text{react}} = \mathbf{1}\{S_{i,t-3:t-1} = 0,\ S_{it} \geq s_0\}, \tag{20}$$

$$S_{it} = \text{monthly purchase spend on the card,} \tag{21}$$

$$B_{it} = \text{monthly card debt or revolving balance.} \tag{22}$$

Here A_{it}^{react} is a reactivation indicator among accounts that were inactive before the reward, S_{it} is the usage outcome, and B_{it} is the balance outcome. These are exactly the three objects a bank needs for a new card: did the account become active, how much spend did it generate, and did it also create debt exposure?

A reduced-form implementation for a randomized or quasi-randomized reward offer can be written as

$$Y_{it} = \alpha_i + \lambda_t + \tau\,\text{Reward}_{it} + \eta' X_{it} + \epsilon_{it}, \tag{23}$$

where Y_{it} may be purchase spend, revolving debt, or an indicator for reactivation among previously inactive cardholders. The estimand is

$$\tau_Y = \mathbb{E}[Y_{it}(r = 1) - Y_{it}(r = 0)], \tag{24}$$

with r denoting the reward treatment. In a bank implementation this should be estimated heterogeneously,

$$\tau_Y(x, r) = \mathbb{E}[Y_i(r) - Y_i(0) \mid X_i = x], \tag{25}$$

where X_i includes prior inactivity, existing card holdings, pre-period merchant mix, line utilization, relationship depth, channel, credit quality, and macro state. Equation (25) is the bridge from Agarwal et al. [1] to the later model: it gives a segment-level reward response surface rather than a single portfolio average.

The output of this block should feed the NPV simulator in four places:

$$\Pr(Z_{i,t+1} = \text{active} \mid Z_{it}, X_{it}, a) \leftarrow \text{activation/reactivation effect,} \tag{26}$$

$$\mathbb{E}[S_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{spend effect,} \tag{27}$$

$$\mathbb{E}[B_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{debt/balance effect,} \tag{28}$$

$$C_{i,t+1}^{\text{reward}}(a) = q_a S_{i,t+1} + F_a + R_a^{\text{signup}}, \tag{29}$$

where q_a is the earn rate or reward cost per dollar, F_a is fixed program cost, and R_a^{signup} is signup or statement-credit cost. The promotion is attractive only if the incremental contribution implied by (26)–(29) is positive after cannibalization and losses. This is the concrete way to avoid importing a headline reward effect as if it were a universal activation parameter.

Behavioral evidence also suggests that payment instruments can affect purchase behavior. Prelec and Simester [25] find in experimental settings that willingness to pay can be higher under credit-card payment than cash payment. For a bank NPV model, this is mechanism evidence: payment method may change the purchase decision, not simply route a fixed purchase through a different rail. The result is not a portfolio-level NPV estimate and should not be used as a direct calibration without issuer data.

The conservative model use is a mechanism prior for the ticket-size component. Let N_{ijt} be transaction count in category j and $\bar{s}_{ijt}$ average ticket size. Then

$$S_{ijt} = N_{ijt}\bar{s}_{ijt}, \tag{30}$$

$$\log \bar{s}_{ij,t+1} = \mu_i + \kappa_j + \beta' X_{it} + \theta_1 \text{CreditCard}_{ij,t+1} + \theta_2 \text{RewardSalience}_{ij,t+1} + \epsilon_{ij,t+1}. \tag{31}$$

The Prelec and Simester [25] result supports allowing θ_1 or θ_2 to differ from zero, but the bank must estimate those parameters with transaction data. This block is explanatory unless validated on issuer data; it should not be a hard-coded multiplier in NPV.

3.3 Cannibalization and incremental usage

For existing clients, the central challenge is cannibalization. A newly issued card can create genuinely incremental spend, shift spend from another issuer, shift spend from the bank's existing card, move checking/debit activity onto credit, or induce balances whose loss and reward costs exceed revenue. To make this estimable, define the following observed spend aggregates over horizon H:

$$S_{iH}^{\text{new}} = \text{spend on the newly issued card}, \tag{32}$$

$$S_{iH}^{\text{oldbank}} = \text{spend on the bank's pre-existing cards}, \tag{33}$$

$$S_{iH}^{\text{debit}} = \text{bank-observed debit/checking purchase activity}, \tag{34}$$

$$S_{iH}^{\text{bank}} = S_{iH}^{\text{new}} + S_{iH}^{\text{oldbank}}. \tag{35}$$

When outside-card spend is not observed, competitor-card substitution is latent and must be inferred from bureau/payment-network aggregates, wallet estimates, or experimental lift in total bank spend. The estimands are:

$$\tau_i^{\text{new card spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{new}}(a) - S_{i,t:t+H}^{\text{new}}(0) \mid \mathcal{I}_{it}], \tag{36}$$

$$\tau_i^{\text{bank spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{all bank cards}}(a) - S_{i,t:t+H}^{\text{all bank cards}}(0) \mid \mathcal{I}_{it}], \tag{37}$$

$$\text{Cannibalization}_i(a) = \tau_i^{\text{new card spend}}(a) - \tau_i^{\text{bank spend}}(a). \tag{38}$$

Equation (36) can be positive while Equation (37) is small or even negative. That is why new-card activation and new-card spend are not sufficient promotion criteria.

In a randomized campaign, these are directly estimable by treatment-control differences:

$$\widehat{\tau}^{\text{new}}(x) = \mathbb{E}[S_{iH}^{\text{new}} \mid A_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{new}} \mid A_i = 0, X_i = x], \tag{39}$$

$$\widehat{\tau}^{\text{bank}}(x) = \mathbb{E}[S_{iH}^{\text{bank}} \mid A_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{bank}} \mid A_i = 0, X_i = x], \tag{40}$$

$$\widehat{\kappa}(x) = 1 - \frac{\widehat{\tau}^{\text{bank}}(x)}{\widehat{\tau}^{\text{new}}(x)} \quad \text{when } \widehat{\tau}^{\text{new}}(x) > 0. \tag{41}$$

Here $\widehat{\kappa}(x)$ is the estimated cannibalization rate from the bank's own existing-card book. If $\widehat{\kappa}(x)$ is close to one, most new-card spend is shifted from old bank cards; if it is close to zero, new-card spend is mostly incremental to the bank. If it is negative, the new card may also lift old bank card spend, for example through relationship deepening.

The latent competitor-transfer component can be represented as

$$\tau_i^{\text{competitor shift}}(a) = \tau_i^{\text{bank spend}}(a) - \tau_i^{\text{debit shift}}(a) - \tau_i^{\text{new consumption}}(a), \tag{42}$$

The debit-shift term can sometimes be estimated from the bank's checking and debit data. The new-consumption term is harder to identify without network or merchant-panel data. Equation (42) should therefore be treated as a decomposition with an uncertainty interval, not as a precisely observed quantity.

Randomized campaign holdouts are the cleanest way to separate organic propensity from treatment effect. When randomization is unavailable, the causal-inference and uplift literatures provide tools, but they do not remove the need for credible identification. The propensity-score framework of Rosenbaum and Rubin [27], true-lift marketing models such as Lo [22], causal forests [34], and double/debiased machine learning [11] are useful methodological references. In a regulated bank environment, they should be used with overlap checks, sensitivity analysis, segment calibration, and careful model-governance review.

For later modeling, the cleanest operational implementation is to estimate a conditional average treatment effect for each downstream outcome,

$$\tau_Y(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x], \tag{43}$$

where Y may be activation, first purchase, first-90-day new-card spend, all-bank-card spend, revolving balance, loss, or full NPV. The causal/uplift module should feed the simulation engine by replacing raw predictions $\mathbb{E}[Y_i \mid A_i = 1, X_i]$ with counterfactual contrasts $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i]$. If randomization is not available, the model must record the identification assumption and cannot be promoted solely on predictive fit.

The handoff to Equation (3) is explicit. Let r_j^{int} be net interchange by merchant category, $q_j(a)$ reward cost by category and offer, m_{it}^{finance} net finance margin on balances, and ℓ_{it} expected loss. Then a horizon-H incremental contribution block is

$$\Delta\Pi_{iH}(a) = \sum_j \{r_j^{\text{int}} - q_j(a)\}\widehat{\tau}_{ijH}^{\text{bank}}(a) + \widehat{\tau}_{iH}^{\text{finance}}(a) - \widehat{\tau}_{iH}^{\text{loss}}(a) - C_i^{\text{campaign}}(a) - C_i^{\text{servicing}}(a), \tag{44}$$

$$\Delta\text{NPV}_i(a) = \sum_{h=0}^{H} \delta_h \, \mathbb{E}[\Delta\Pi_{ih}(a) \mid \mathcal{I}_{i0}]. \tag{45}$$

The key design choice is that interchange and reward revenue use $\widehat{\tau}^{\text{bank}}$, not raw new-card spend S^{new}. This is how the cannibalization model changes the optimization target.

4 Lifetime Expenditure and Customer Lifetime Value

4.1 Customer-base models

The third requested problem is how to model lifetime expenditure of a credit card user. Marketing science provides a natural starting point. Schmittlein et al. [30] develop a model for inferring which customers are still active and what they will do next from transaction timing and frequency. This is directly relevant for card accounts because a customer can silently become inactive before formal closure.

Fader et al. [12] propose the BG/NBD model as a tractable alternative to the Pareto/NBD model for predicting repeat-purchase behavior. Fader et al. [13] connect recency, frequency, and monetary value to customer lifetime value. These models are useful baselines for transaction frequency, active/dormant status, and spend per transaction. They are especially relevant for transactors whose value is driven primarily by purchase volume and interchange rather than finance charges.

Relationship duration is another core object. Bolton [9] models the duration of a customer's relationship with a continuous service provider. Credit cards have similar duration dynamics: fees, service quality, fraud events, line changes, disputes, rewards devaluations, and pricing changes can alter retention, dormancy, and usage.

Table 2: How Section 3 evidence should enter a bank model.

Literature	Mathematical object	Model use
Koulayev et al. [20]	Structural adoption and conditional use of payment instruments.	Separate opened/activated state from conditional card-use choice; include competing payment instruments and transaction context.
Agarwal et al. [1]	Treatment effect of rewards on spend, debt, and reactivation.	Estimate $\tau_Y(x,r)$ for reactivation, spend, and balances; feed the activation transition, conditional spend, balance, and reward-cost blocks in Equations (26)–(29).
Prelec and Simester [25]	Experimental effect of payment medium on willingness to pay.	Use as mechanism support for the ticket-size equation (31); estimate the payment-medium coefficient on issuer transaction data before using it in NPV.
Uplift/CATE literature	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$ for activation, spend, balance, loss, and NPV.	Estimate new-card lift, all-bank-card lift, and cannibalization in Equations (39)–(41); feed incremental contribution through Equation (44).

4.2 Why card NPV is broader than plain CLV

Credit cards are not a pure repeat-purchase setting. A bank needs the lifetime path of at least six linked processes:

$$
\begin{aligned}
S_{it} &= \text{purchase spend and transaction count,} && (46)\\
B_{it} &= \text{statement and revolving balance,} && (47)\\
P_{it} &= \text{payment amount and payment rate,} && (48)\\
D_{it} &= \text{delinquency/default state,} && (49)\\
U_{it} &= \text{active, dormant, closed, or charged-off usage state,} && (50)\\
Q_{it} &= \text{rewards, fraud, disputes, servicing, and other costs.} && (51)
\end{aligned}
$$

These processes are jointly determined. High spend can be profitable when it is durable and low loss, but low value when driven by high rewards cost or low-margin categories. Revolving balances can raise interest income and expected loss simultaneously. A durable active customer may be unprofitable if servicing, fraud, or rewards costs dominate revenue.

Thus, a bank should treat CLV models as transaction and duration components of a broader card-NPV engine. Lifetime expenditure is necessary but not sufficient: the model must translate expenditure into interchange, rewards cost, balances, finance charges, losses, capital, and servicing cost.

5 Recommended Modeling Architecture

The literature supports a modular state-transition architecture. Let X_{it} collect customer attributes, internal relationship history, bureau variables, card terms, macro variables, and campaign information. Let Z_{it} denote the latent card state:

$$Z_{it} \in \{\text{prospect}, \text{opened}, \text{activated}, \text{first-use}, \text{active}, \text{dormant}, \text{delinquent}, \text{closed}, \text{charged-off}\}.$$

A practical model can be written as

$$\Pr(Z_{i,t+1} = z' \mid Z_{it} = z, X_{it}, a), \tag{52}$$

$$\mathbb{E}(S_{i,t+1}, B_{i,t+1}, P_{i,t+1}, D_{i,t+1}, Q_{i,t+1} \mid Z_{i,t+1}, X_{it}, a). \tag{53}$$

Equations (52) and (53) can be implemented as a hurdle model, survival model, dynamic panel, machine-learning model, structural dynamic discrete-choice model, or simulation engine. The choice depends on data scale, interpretability, regulatory constraints, and the decision horizon. The dynamic discrete-choice tradition, exemplified by Rust [28], is useful conceptually because account states and bank actions evolve over time, although a full structural model may be unnecessary for many targeting applications. Dynamic panel methods such as Blundell and Bond [6] provide another reference point when lagged outcomes and persistent customer effects are central.

The architecture should contain the following modules:

1. **Eligibility and offer module.** Forecast approval, line, APR, annual fee, reward terms, disclosures, and compliance constraints.

2. **Causal response module.** Estimate incremental open, activation, first-use, and wallet-share effects of the campaign.

3. **Activation and early-usage module.** Forecast activation, digital-wallet provisioning, first purchase, first 90-day spend, and early dormancy.

4. **Spend and category module.** Forecast purchase volume, transaction count, ticket size, merchant-category mix, and seasonality.

5. **Balance and payment module.** Forecast revolving behavior, promotional balance-transfer behavior, payment rate, and interest revenue.

6. **Risk and loss module.** Forecast delinquency, default, exposure at default, loss given default, fraud, disputes, and recovery.

7. **Attrition and dormancy module.** Forecast inactivity, closure, product switching, retention response, and charge-off exit.

8. **Finance module.** Convert simulated paths into discounted contribution after rewards, servicing, funding, capital, tax/accounting treatment, and acquisition cost.

The output should be a distribution for $\Delta\text{NPV}_i(a)$, not only a point estimate. Marketing decisions should consider expected value, uncertainty, downside tail risk, portfolio concentration, and operational constraints. A portfolio optimizer may rank customers by expected incremental NPV subject to risk, fairness, capacity, and regulatory constraints.

6 Component Decomposition for a Bank-Grade Card NPV Engine

The proposed decomposition into losses, balances, pre-provision net revenue (PPNR), and portfolio-management overlays is directionally right. It resembles how banks already organize credit-card forecasting in risk, finance, and stress testing: expected credit loss is commonly decomposed into probability of default (PD), loss given default (LGD), and exposure at default (EAD); account-level balances and usage drive both revenue and exposure; and PPNR converts balances, rates, fees, expenses, and relationship effects into pre-loss operating profit. The Basel credit-risk framework formalizes the PD/LGD/EAD vocabulary for regulatory capital [4]. U.S. stress-testing practice also separates loan losses from PPNR, while forecasting portfolio balances and revenues under macro scenarios [7, 8]. Accounting standards for current expected credit losses require lifetime expected-loss measurement based on historical experience, current conditions, and reasonable and supportable forecasts [14]. These references do not dictate a marketing NPV model, but they support the decomposition as a bank-native control structure.

The classification needs three amendments before it can serve as an NPV optimization architecture. First, it should separate *state variables* from *cash-flow identities*. Purchases, cash advances, payments, lines, promotional balances, delinquency, and inactivity are state or flow variables; interest income, interchange, rewards cost, fraud, servicing, and credit losses are cash-flow consequences of those states. Second, it should separate *forecasts* from *treatment effects*. A high forecast of spend on the new card is not the same as incremental bank spend after cannibalization, as Section 3 emphasized. Every component must therefore produce $Y_i(a) - Y_i(0)$, not only $Y_i(a)$. Third, it should add several missing economic items: interchange and merchant-category economics, rewards and signup-bonus liability, fraud and dispute losses, servicing and collections expense, funding and liquidity cost, capital cost, tax/accounting treatment, fair-lending and adverse-action constraints, and relationship spillovers to deposits or other products.

6.1 Loss Module: PD, LGD, and EAD

The loss block is the most standard part of the proposed taxonomy. Credit scoring and behavioral scoring literatures model the probability that a borrower or account will default or enter delinquency using application, behavioral, bureau, and macro variables [17, 21, 32]. For a card account, a horizon-h expected-loss identity is

$$EL_{i,t,h}(a) = PD_{i,t,h}(a)\, LGD_{i,t,h}(a)\, EAD_{i,t,h}(a), \tag{54}$$

with the incremental object

$$\Delta EL_{i,t,h}(a) = \mathbb{E}[EL_{i,t,h}(a) - EL_{i,t,h}(0) \mid \mathcal{I}_{it}]\,. \tag{55}$$

Equation (54) is an accounting/risk identity, not an independence assumption. In credit cards, PD, LGD, and EAD are linked through utilization, credit line, payment behavior, delinquency state, collections treatment, and macro conditions. A line increase may raise activation and spend, but it can also raise utilization and EAD. A balance-transfer promotion may raise near-term balances and interest income, but it changes both the timing of EAD and the risk mix of accounts.

The PD component should therefore be a transition or hazard model, not a static score detached from the card lifecycle:

$$\Pr(D_{i,t+h} = 1 \mid Z_{it}, X_{it}, B_{it}, P_{it}, L_{it}, M_{gt}, a), \tag{56}$$

where D is default or charge-off, Z is lifecycle state, B balance, P payment behavior, L line, and M macro state. Months-on-book and vintage should enter this equation because newly originated accounts season differently from mature accounts. That point is operationally important for campaign NPV: an apparently low-loss first-six-month window may simply precede the period in which a vintage reaches peak delinquency.

LGD should be modeled as a recovery and charge-off severity process conditional on default:

$$LGD_{i,t,h} = 1 - \frac{\mathbb{E}[\text{Recoveries}_{i,t:t+h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}{\mathbb{E}[EAD_{i,t,h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}. \tag{57}$$

The LGD literature emphasizes that recovery rates vary with collateral, seniority, macro state, and workout environment [31]; unsecured credit-card LGD must additionally account for collections strategy, bankruptcy, hardship programs, and sale of charged-off debt. For a marketing NPV engine, LGD is usually less sensitive to the offer than EAD and PD, but it is not constant across macro states or customer segments.

EAD must be tied to the balance and line module:

$$EAD_{i,t,h}(a) = B^{\text{principal}}_{i,t+h}(a) + B^{\text{nonprincipal}}_{i,t+h}(a) + \text{Accrued}_{i,t+h}(a) + \text{UndrawnDraw}_{i,t,h}(a). \tag{58}$$

This is where the user's proposed "principal versus non-principal balances and losses" belongs. The model should maintain a balance subledger because purchase principal, cash advances, balance transfers, fees, interest, and promotional balances have different APRs, payoff rules, recoverability, revenue recognition, consumer behavior, and regulatory treatment. Rolling them into one balance can make a promotion look profitable by mixing low-margin teaser balances with high-margin revolving balances or by assigning losses to the wrong balance type.

6.2 Balance, Line, and Usage Module

The balance block is the mechanical bridge between usage, revenue, and loss. At monthly frequency, a card account should satisfy a stock-flow identity such as

$$\begin{aligned} B_{i,t+1} = B_{it} + S^{\text{purchase}}_{it} + S^{\text{cashadv}}_{it} + BT_{it} + Fee_{it} + Int_{it} \\ - Pay_{it} - Credit_{it} - Chargeoff_{it} - OtherAdj_{it}, \end{aligned} \tag{59}$$

where purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments are separately retained. This identity is not optional in a production NPV model: without it, the model can forecast sales volume, balances, interest income, and losses that are mutually inconsistent.

The behavioral forecasting pieces around the identity are:

$$\Pr(Z_{i,t+1} \mid Z_{it}, X_{it}, a), \tag{60}$$

$$\mathbb{E}[S^{\text{purchase}}_{ij,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a], \tag{61}$$

$$\mathbb{E}[Pay_{i,t+1} \mid B_{it}, S_{it}, Z_{it}, X_{it}, M_{gt}, a], \tag{62}$$

$$\mathbb{E}[L_{i,t+1} \mid L_{it}, X_{it}, B_{it}, D_{it}, a], \tag{63}$$

with utilization $U_{it} = B_{it}/L_{it}$. The household-finance evidence in Section 2 enters (61) and (62); the liquidity and credit-line evidence in Gross and Souleles [16] enters (63); the payment-choice and rewards evidence in Section 3 enters (60) and (61).

The proposed list of balance variables is mostly right but should be amended. "Originations" should be split into application approval, booked account, initial line, activation, and first use. "Sales volume" should be category-level purchase volume, not only total spend, because interchange, rewards, disputes, fraud, and macro cyclicality vary by merchant category. "Lines" should be modeled as a bank action and constraint, not only a descriptive variable. "Cash advance" should be separated from purchases and balance transfers because it has different fees, APR, risk selection, and customer intent. "Interest fees" should be decomposed into finance charges, late fees, annual fees, balance-transfer fees, cash-advance fees, and fee waivers, especially after the CARD Act evidence that regulation and disclosure alter fee economics [2].

6.3 PPNR and Contribution Module

PPNR is a useful finance lens because it separates operating revenue and expense from credit losses. In a card NPV engine, however, it should be defined at account-month level before aggregation:

$$PPNR_{it}(a) = NII_{it}(a) + NIR_{it}(a) - NIE_{it}(a), \tag{64}$$

where

$$\begin{aligned} NII_{it}(a) &= Int_{it}(a) - FundCost_{it}(a), && (65)\\ NIR_{it}(a) &= Interchange_{it}(a) + FeeIncome_{it}(a) + OtherIncome_{it}(a), && (66)\\ NIE_{it}(a) &= RewardCost_{it}(a) + ServicingCost_{it}(a) + FraudLoss_{it}(a)\\ &\quad + DisputeCost_{it}(a) + CollectionCost_{it}(a) + AcqCost_{it}(a). && (67) \end{aligned}$$

The stress-testing literature supports this separation at the bank level [7, 8]; the marketing and CLV literatures support discounting future net contribution [5, 29, 33]. For our purposes, the monthly incremental contribution is

$$\Delta\Pi_{it}(a) = \Delta PPNR_{it}(a) - \Delta EL_{it}(a) - \Delta K_{it}(a) - \Delta Tax_{it}(a), \tag{68}$$

and the customer NPV is the discounted sum of (68), plus any one-time acquisition or conversion costs not already embedded in monthly PPNR.

Deposits should be included, but not as an undifferentiated PPNR line item. For a card-only campaign, deposits are a relationship spillover:

$$\begin{aligned} \Delta RelValue_{it}(a) &= \Delta DepositMargin_{it}(a) + \Delta LoanMargin_{it}^{\text{other}}(a)\\ &\quad + \Delta FeeIncome_{it}^{\text{other}}(a) - \Delta Servicing_{it}^{\text{other}}(a). \end{aligned} \tag{69}$$

This term should be estimated only when the campaign plausibly changes relationship depth, primary-bank status, or deposit behavior. Otherwise, adding deposit PPNR to card NPV risks attributing organic household banking value to a card offer. In the existing-client setting, randomized holdouts are especially important because relationship value is strongly confounded by pre-existing engagement.

6.4 Cost-Cutting and Portfolio-Management Overlays

The user's cost-cutting topics are not a separate objective from NPV; they are control and segmentation layers around the same component engine. Inactivity/attrition should be a state-transition model,

$$\Pr(Z_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid Z_{it}, X_{it}, S_{it}, B_{it}, Service_{it}, a), \tag{70}$$

building on duration and customer-base models [9, 12, 30]. Its NPV role is to shorten or lengthen the cash-flow horizon and to trigger retention, line-management, or closure actions. Cost-cutting based on inactivity is only valid if it does not destroy positive future option value or create adverse selection in the remaining portfolio.

Promotional accounts should be represented by explicit promo states:

$$PromoState_{it} = (\text{APR}_{it}^{promo}, \text{expiry}_{it}, B_{it}^{promo}, \text{reward liability}_{it}). \tag{71}$$

This is necessary because signup bonuses, teaser APRs, balance transfers, and deferred-interest offers create cash-flow timing that differs from ordinary revolving balances. A promo account can

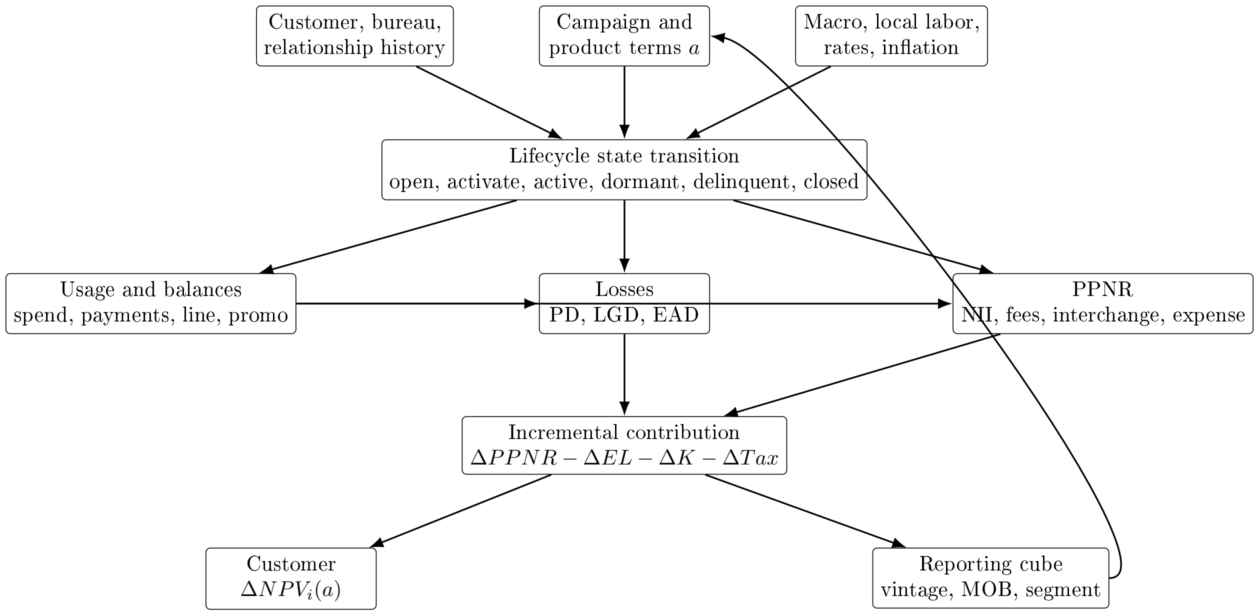

Figure 1: Schematic architecture for a component-based incremental NPV engine.

have high early sales volume and negative NPV if reward and funding costs arrive before durable profitable balances or if post-promo attrition is high.

Product conversion should be modeled as a treatment-induced transition among products:

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, Z_{it}, X_{it}, a), \tag{72}$$

with the incremental value measured against organic conversion. This is the same causal problem as new-card issuance: the bank should not credit the conversion campaign for customers who would have converted anyway, and it should subtract cannibalized annual fees, rewards economics, or portfolio balances from the old product.

Months-on-book and vintage should be required dimensions, not optional reporting labels. They enter PD, spend, payment, attrition, and profitability because accounts season after origination, campaigns are launched in different macro states, and underwriting or pricing policies change across cohorts. A production reporting cube should therefore support segmentation by at least product, channel, campaign, offer terms, vintage, months-on-book, FICO or risk band, geography, macro state, line/utilization band, revolver/transactor status, principal/non-principal balance type, promotional status, relationship depth, and protected-class proxy monitoring where legally appropriate. Flexible segmentation is a reporting and governance requirement; it should not imply that every segment receives a separate unconstrained model. Hierarchical or pooled models are often preferable when segment-level data are sparse.

6.5 Schematic System Design

Figure 1 shows the intended system architecture. The important feature is that customer state, bank action, and macro state feed all behavioral modules; the finance layer then imposes accounting identities and converts simulated paths into incremental cash flows. The reporting layer is downstream of the model, so segmentation can be flexible without changing the definition of NPV.

Table 3: Component outputs required by the incremental NPV model.

Component	Required output	Connection to NPV
Response and activation	$\Delta\,\Pr(\text{open})$, $\Delta\,\Pr(\text{active})$, $\Delta\,\Pr(\text{first use})$	Gates all future spend, balance, expense, and loss paths.
Usage and balances	Incremental purchases, cash advances, balance transfers, payments, lines, promo balances, and utilization	Drives interchange, rewards, finance charges, funding cost, EAD, and capital.
Losses	ΔPD, ΔLGD, ΔEAD, and ΔEL	Subtracted from PPNR as expected credit loss; also informs capital and downside-risk constraints.
PPNR	Incremental net interest income, non-interest income, and non-interest expense	Provides pre-loss contribution before expected loss, capital, tax, and relationship spillovers.
Attrition and conversion	Incremental dormancy, closure, retention, and product-migration paths	Determines cash-flow duration and whether value is new, retained, or cannibalized from another bank product.
Relationship module	Incremental deposit margin and other-product contribution	Added only when credibly caused by the card action, preferably with randomized or quasi-experimental evidence.
Reporting segmentation	Aggregation by product, vintage, MOB, risk, geography, promo, and balance type	Supports governance, monitoring, and cost actions; does not by itself define the optimization target.

6.6 Connection Back to Incremental NPV

The component architecture is useful only if every module returns a counterfactual cash-flow contrast. The final object should be

$$\Delta\text{NPV}_i(a) = -\Delta A_i(a) + \sum_{h=0}^{H} \delta_h\, \mathbb{E}\left[\begin{aligned} \Delta PPNR_{i,t+h}(a) - \Delta EL_{i,t+h}(a) - \Delta K_{i,t+h}(a) \\ - \Delta Tax_{i,t+h}(a) + \Delta RelValue_{i,t+h}(a) \mid \mathcal{I}_{it} \end{aligned}\right], \tag{73}$$

where each Δ is relative to no offer, no conversion, or the relevant business-as-usual action. Table 3 summarizes the handoff from each component to Equation (73).

This framing also clarifies what should be amended in the original classification. Losses, balances, and PPNR are essential, but the production system should explicitly include rewards, interchange, fraud/disputes, servicing and collections, capital and funding, taxes/accounting, relationship spillovers, and causal cannibalization. The model should preserve the principal/non-principal subledger and vintage/months-on-book dimensions because they affect both economics and governance. Finally, cost-cutting decisions should be evaluated as actions inside the same NPV system rather than as external heuristics; an inactive account, a promotional account, or a product conversion is valuable only to the extent that its incremental discounted contribution is positive after losses, costs, and cannibalization.

7 Decision and Measurement Pitfalls in Card NPV

The consultant concerns are well founded. They are not objections to using NPV; they are reasons the NPV system must be explicitly staged, causal, and uncertainty-aware. The marketing and CLV literatures support a discounted contribution objective [5, 10, 29, 33], but the card setting adds long-horizon credit, funding, capital, and account management dynamics. The right response is not to replace NPV with response, approval, or sales volume. It is to make clear which NPV is being estimated, for which decision, on which population, and under which downstream policy.

7.1 NPV Is a Long-Horizon Latent Quantity

Customer NPV is unintuitive because it is not directly observed. The bank observes partial cash flows over finite calendar time; it does not observe the counterfactual no-offer path, the customer's full outside-wallet behavior, or the terminal value of the relationship. A more honest notation is therefore

$$\Delta\mathrm{NPV}_i(a;\theta,\pi) = \sum_{h=0}^{H} \delta_h \mathbb{E}_\theta[\Delta CF_{i,t+h}(a,\pi) \mid \mathcal{I}_{it}] + \delta_H \mathbb{E}_\theta[\Delta TV_{i,t+H}(a,\pi) \mid \mathcal{I}_{it}], \tag{74}$$

where θ collects model parameters and scenario assumptions, π is the future account-management policy, CF is the component cash flow from Equation (73), and TV is terminal value. This notation makes the instability visible: rankings can change when assumptions about spend decay, revolve rate, attrition, default timing, recovery, funding cost, capital cost, or terminal value change.

A bank implementation should therefore report not only $\widehat{\Delta\mathrm{NPV}}_i(a)$, but also a sensitivity or posterior distribution:

$$\left\{\Delta\mathrm{NPV}_i(a;\theta^{(s)},\pi^{(s)})\right\}_{s=1}^{S}. \tag{75}$$

For campaign selection, the robust decision statistic may be expected incremental NPV, lower-tail NPV, probability of negative NPV, or expected NPV subject to capital and loss constraints. This is consistent with the stress-test view that balance, revenue, and loss paths should be evaluated under macro scenarios [7, 8]. It also prevents proxy metrics such as first-90-day spend from silently becoming promotion criteria for lifetime value.

7.2 Card Value Is Behaviorally Heterogeneous

The second issue is that card profitability is not one-dimensional. Two customers with the same first-month spend can have opposite economics if one is a low-risk transactor with high interchange and modest rewards cost, while the other revolves promotional balances, pays slowly, uses cash advances, and later charges off. The appropriate object is a heterogeneous treatment effect over the full cash-flow vector:

$$\tau_V(x,a) = \mathbb{E}[V_i(a) - V_i(0) \mid X_i = x], \tag{76}$$

where V_i is not one outcome but the discounted contribution generated by spend, balances, payments, losses, rewards, fraud, servicing, funding, and capital. Uplift and causal-forest methods are useful for estimating heterogeneous treatment effects [11, 22, 34], and statistical treatment-rule work emphasizes choosing actions by expected welfare or payoff under heterogeneity rather than by average effect alone [23].

This implies that the model should not rank customers by a single revenue or risk proxy. A useful diagnostic decomposition is

$$\begin{aligned}\tau_V(x,a) = \tau_{\text{interchange}}(x,a) + \tau_{\text{finance}}(x,a) + \tau_{\text{fees}}(x,a) - \tau_{\text{rewards}}(x,a) \\ - \tau_{\text{servicing}}(x,a) - \tau_{\text{fraud}}(x,a) - \tau_{\text{loss}}(x,a) - \tau_{\text{funding/capital}}(x,a).\end{aligned} \tag{77}$$

The decomposition need not be estimated by separate independent models; indeed, the components are linked through balances and states. But it should be reported, because a high average NPV segment can hide subsegments whose value comes from fragile assumptions about revolve, funding, or losses.

7.3 The Evaluated Population Changes Through the Funnel

The population changes as the customer moves from prospect to applicant to approved account to booked account to active user. This is a selection problem: the covariates observed, the eligible action set, and the conditional value distribution all change after each gate. The sample-selection literature is the classical warning that outcome models estimated only on selected observations need not describe the original population [18]. In the card funnel, the same issue appears in operational form:

$$\mathcal{P}_0 = \{i : \text{marketable prospects}\}, \tag{78}$$

$$\mathcal{P}_1 = \{i \in \mathcal{P}_0 : \text{respond or apply}\}, \tag{79}$$

$$\mathcal{P}_2 = \{i \in \mathcal{P}_1 : \text{approved and line assigned}\}, \tag{80}$$

$$\mathcal{P}_3 = \{i \in \mathcal{P}_2 : \text{booked and activated}\}. \tag{81}$$

The right estimand changes with the conditioning set. For example,

$$\tau^{\text{target}}(x) = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_0, X_i = x], \tag{82}$$

$$\tau^{\text{approve}}(x) = \mathbb{E}[\text{NPV}_i(\text{approve}) - \text{NPV}_i(\text{decline}) \mid i \in \mathcal{P}_1, X_i = x], \tag{83}$$

$$\tau^{\text{usage}}(x) = \mathbb{E}[\text{NPV}_i(\text{line, terms}) - \text{NPV}_i(\text{baseline}) \mid i \in \mathcal{P}_2, X_i = x]. \tag{84}$$

These are not interchangeable. A targeting model can use prospect and relationship data but not application-only variables. An underwriting model can use application and bureau data that were unavailable at prospect selection. An active-account management model can use payment, utilization, transaction, service, and delinquency history that do not exist for non-booked prospects. Therefore, a bank should maintain separate model cards, validation populations, and data-availability timestamps for each funnel stage.

7.4 The Counterfactual and Objective Change by Decision

The fourth issue is that "the NPV objective" is not one universal treatment contrast. It is a family of decision-specific contrasts. Direct-marketing selection asks whether contact has positive incremental value net of contact cost [10]. Portfolio budget allocation asks whether card marketing beats redeploying the same budget to another product, channel, or retention program. Underwriting asks whether approval creates value relative to decline, subject to credit policy, fairness, compliance, and risk appetite. Line assignment asks which line maximizes value over a feasible menu while respecting loss and capital constraints. Formally,

$$a_d^*(x) \in \arg\max_{a \in \mathcal{A}_d(x)} \mathbb{E}[U_d\{\Delta\text{NPV}_i(a), Risk_i(a), Cap_i(a), Fair_i(a)\} \mid X_i = x, \mathcal{P}_d], \tag{85}$$

Table 4: Decision-specific counterfactuals for card NPV.

Decision	Population	Counterfactual	Objective
Targeting	Marketable prospects or existing clients	Contact versus suppress	Incremental NPV net of campaign cost.
Budget allocation	Eligible campaign universe	Card campaign versus redeploy budget	Portfolio NPV per scarce marketing dollar or capacity unit.
Underwriting	Applicants or responders with application data	Approve versus decline, or approve with terms versus decline	Risk-adjusted NPV subject to policy and compliance constraints.
Line assignment	Approved or preapproved accounts	Line amount ℓ versus alternative line amounts	NPV over line menu, accounting for usage, EAD, losses, and capital.
Account management	Booked accounts	Maintain, reprice, retain, convert, line change, or close	Forward NPV under updated behavior and risk state.

where d indexes the decision stage and $\mathcal{A}_d(x)$ is the feasible action set at that stage. The utility U_d may be expected NPV for low-risk targeting, constrained NPV for underwriting, or portfolio NPV subject to budget, capacity, capital, and fairness constraints.

Table 4 gives the operational version. The main governance point is that a model validated for one row should not be reused for another row without re-estimating the counterfactual and checking that the conditioning population and action set match.

7.5 Decisions Are Interdependent Over Time

Finally, decisions are interdependent. Acquisition changes the population that underwriting sees. Underwriting and line assignment change the balance, utilization, and risk states that account management later sees. Future line, pricing, collections, retention, or conversion policies change the lifetime economics assumed at acquisition. The dynamic treatment-regime and Markov decision-process literatures provide the natural language for this problem: the value of an action depends on the future policy that follows it [24, 26, 28].

A compact representation is

$$Z_{i,t+1} \sim p_\theta(Z_{i,t+1} \mid Z_{it}, X_{it}, a_t), \tag{86}$$

$$CF_{it} = c_\theta(Z_{it}, X_{it}, a_t), \tag{87}$$

$$\pi_t : (Z_{it}, X_{it}) \mapsto a_t, \tag{88}$$

with policy value

$$V_i^\pi(z,x) = \mathbb{E}_\theta^\pi \left[\sum_{h=0}^{H} \delta_h CF_{i,t+h} + \delta_H TV_{i,t+H} \mid Z_{it} = z, X_{it} = x \right]. \tag{89}$$

The acquisition model should therefore state its assumed downstream policy π: for example, line-increase rules, repricing rules, collections policy, retention offers, dormancy closure, and product-conversion policy. If those policies change, the upstream acquisition NPV should be recalibrated. This is especially important for cost-cutting initiatives: closing inactive accounts, reducing promotional offers, tightening lines, or converting products may improve near-term expense or losses while reducing the value of cohorts that were acquired under older assumptions.

The practical recommendation is to treat card NPV as a policy-evaluation system, not a one-time score. Each major decision should store the population, available variables, action set, counterfactual, downstream policy assumption, cash-flow decomposition, and sensitivity range. That metadata is not bureaucracy; it is what prevents an upstream model from optimizing a downstream world that the bank no longer operates.

8 Empirical Design and Validation

8.1 Data requirements

A bank-grade data set should join campaign exposure, channel, creative, offer terms, application, underwriting, activation, card transactions, balances, payments, delinquency, charge-off, recoveries, fraud, disputes, servicing, relationship products, deposits, payroll indicators, bureau refreshes, and local macro series. The data should preserve timing. Variables measured after campaign exposure must not be used as if they were available at targeting time.

For the existing-client new-card problem, the data must also identify cannibalization. At minimum, the bank should measure whether new-card spend comes from:

a. genuinely incremental customer spend;

b. spend shifted from a competitor card;

c. spend shifted from the bank's existing card;

d. debit or deposit activity shifted onto credit;

e. temporary promotion-induced spend that decays after the offer.

Only the first two categories are necessarily positive for the issuing bank, and even those must be netted against rewards, losses, and costs.

8.2 Promotion criteria

The primary promotion criterion should be randomized or credibly quasi-experimental lift in incremental NPV. Response AUC, approval AUC, activation lift, first-purchase lift, and first-90-day spend are useful diagnostics, but they are proxy metrics. They should nominate models for further testing rather than establish lifetime NPV validity.

Recommended validation splits include:

- time-based holdouts for economic drift;
- campaign randomized holdouts for treatment effects;
- vintage holdouts for lifetime trajectories;
- macro-stress backtests around unemployment, rate, and inflation shocks;

- segment calibration across score bands, income proxies, geography, tenure, relationship depth, channel, and product;
- counterfactual cannibalization analysis for existing-client offers.

9 Open Gaps and Reviewer Risks

The public literature does not fully identify issuer-specific new-card usage and incremental NPV. Payment-choice and rewards studies show mechanisms, but the activation probability for a particular bank, product, offer, channel, and customer segment is a proprietary empirical object. Likewise, CLV models provide powerful baselines for transaction frequency and duration, but bank NPV requires integration with expected credit loss, rewards, fraud, funding, capital, and regulatory constraints.

The largest reviewer risks are:

1. treating activation or first spend as lifetime NPV evidence;
2. treating raw propensity as causal lift;
3. ignoring cannibalization from the bank's existing cards;
4. estimating spend without macro and employment state dependence;
5. separating marketing CLV from credit risk and balance-sheet cost;
6. using public rewards estimates as universal program parameters;
7. reusing one NPV model across targeting, underwriting, line assignment, and account management without changing the counterfactual;
8. validating on booked or active accounts while applying the result to prospects or applicants without correcting for funnel selection.

10 Source-Support Audit

This paper is a structured literature survey for product and model-design purposes. It is not yet a complete model-governance literature audit. Primary source pages, abstracts, and working-paper summaries were inspected on June 26–27, 2026. Full-text technical sections, appendices, citation-count metadata, retraction checks, and full backward/forward snowballing remain open. Table 5 records how the main sources are used.

Table 5: First-pass source-support ledger.

Source	Classification	Use in this paper
Bult and Wansbeek [10]	Direct marketing optimization	Supports the claim that campaign selection is normally an expected-profit problem, not a response-probability problem alone.

Source	Classification	Use in this paper
Berger and Nasr [5], Rust et al. [29], and Venkatesan and Kumar [33]	CLV and customer-equity foundations	Support the discounted-contribution framing; the paper's credit-card NPV equation is a bank-specific extension with loss, capital, funding, fraud, rewards, and cannibalization terms.
Ganong and Noel [15]	Household finance	Supports the claim that labor-income and unemployment events affect high-frequency spending and should enter spend forecasts.
Johnson et al. [19]	Income-shock evidence	Supports the claim that liquidity and transfer timing can produce heterogeneous short-run spending responses.
Gross and Souleles [16]	Credit-card account evidence	Supports the claim that credit limits and interest rates affect card debt behavior and interact with liquidity constraints.
Agarwal et al. [2]	Credit-card regulation evidence	Supports the claim that fees, disclosures, and payment nudges are material NPV components and governed levers.
Baker et al. [3]	Macro-shock evidence	Supports the claim that major shocks can alter spending levels and category composition.
Koulayev et al. [20]	Payment-choice model	Supports the distinction between payment-instrument adoption and use.
Agarwal et al. [1]	Rewards and usage evidence	Supports the claim that rewards can shift card spend and debt; not used as a universal activation-rate estimate.
Prelec and Simester [25]	Behavioral mechanism	Supports the mechanism that payment method can affect purchase behavior; not used for portfolio calibration.
Schmittlein et al. [30], Fader et al. [12], and Fader et al. [13]	CLV foundations	Support use of transaction-frequency, recency, monetary-value, and active status models as components of card lifetime modeling.
Bolton [9]	Duration model	Supports use of relationship-duration or hazard components for attrition and dormancy.
Basel Committee on Banking Supervision [4]	Regulatory capital framework	Supports PD/LGD/EAD terminology and the use of exposure, default probability, and severity as separate risk components; not a marketing NPV optimization paper.
Financial Accounting Standards Board [14]	Accounting standard	Supports lifetime expected-credit-loss framing under current conditions and reasonable forecasts; not used to define customer-level causal lift.
Board of Governors of the Federal Reserve System [7] and Board of Governors of the Federal Reserve System [8]	Supervisory stress-test methodology	Support bank-level separation of projected losses, balances, revenues, and PPNR under macro scenarios; used here as architecture support, not as a customer-acquisition estimator.

Source	Classification	Use in this paper
Hand and Henley [17], Thomas [32], and Lessmann et al. [21]	Credit-scoring surveys and benchmark	Support use of application and behavioral scoring methods for PD and delinquency/default components; not used to choose one production model.
Schuermann [31]	LGD survey	Supports modeling LGD as a severity/recovery object distinct from PD and EAD, with macro and workout-state dependence.
Rosenbaum and Rubin [27], Lo [22], Wager and Athey [34], and Chernozhukov et al. [11]	Causal/uplift methods	Support methodological framing for treatment effects; not bank-specific evidence.
Heckman [18]	Sample-selection foundation	Supports the warning that models estimated after response, approval, or booking need not describe the original prospect population.
Manski [23]	Statistical treatment rules	Supports choosing actions by expected payoff under heterogeneous effects, rather than ranking by a single average response or risk measure.
Puterman [26], Murphy [24], and Rust [28]	Dynamic decision and policy-value methods	Support representing acquisition, underwriting, line assignment, and account management as linked sequential decisions whose value depends on downstream policy.

No paper in this survey is used to conclude a universal activation rate, spend elasticity, or NPV parameter. The activation and cannibalization parameters for a large U.S. retail bank should be estimated from internal experiments or credible quasi-experiments.

11 Conclusion

Estimating the NPV of a new credit card customer requires a joint model of marketing response, payment-instrument use, lifetime spend, revolving balances, risk, cost, and macro state dependence. The literature provides strong evidence for several components: income and employment affect spending, liquidity and credit terms affect card debt, adoption differs from use, rewards can change usage, and CLV models can forecast transaction and duration components. It does not provide a universal public formula for whether an existing bank client will use a particular new card or what that customer's incremental NPV will be. That object is bank-, product-, campaign-, and state-specific. A defensible bank model should therefore combine public-literature mechanisms with internal randomized or quasi-experimental lift measurement and should promote models on incremental risk-adjusted NPV rather than raw response or spend propensity.

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